

A New Survival and Prognosis Predictive Model for Combined-Small Cell Lung Cancer (C-SCLC): A Machine Learning Approach

Parzon Eyzadpur Faridani¹ and Kaijie Yu¹

¹Claremont Graduate University, Claremont, California, USA
parzon-eyzadpur.faridani@cgu.edu; kaijie.yu@cgu.edu

Abstract

Combined Small Cell Lung Cancer (C-SCLC), a rare variant subtype of lung cancer has its distinct characteristics and prognosis challenges. Despite its clinical significance, there exists a knowledge gap in the diagnosis, treatment, and prognosis of C-SCLC. Utilizing the SEER database, an authoritative source of cancer data in the United States, we applied advanced machine learning techniques to analyze the overall survival (OS) of C-SCLC patients from 2004 to 2020, across multiple staging systems. Through rigorous data preprocessing and analysis, we developed predictive models that highlight the prognosis factors of C-SCLC but also underscore the subtype's important features and analysis across stages. Our findings provide significant insights and contributions into the survival outcomes of C-SCLC patients, for their distinction within lung cancer classifications. Our study of C-SCLC falls among the very few longitudinal studies and analysis. We presented a model to predict the OS (Overall Survival) for patients with this rare subtype from the year 2004 to 2015, using the American Joint Committee on Cancer (AJCC) 6th edition, alongside visualizations from the analysis, and the code to reproduce these results. We only use the 2004-2015 data to model to ensure consistency in the corresponding staging system (AJCC 6th Edition). We presented a model with 81% recall for patients at high risk (less than 9 months survival) with the most contributing factors to this prediction, which are Metastasis, Chemotherapy, Radiation, Surgery and Tumor Size.

1. Introduction

Lung cancer, recognized as one of the primary causes of cancer-associated mortality globally (Schabath and Cote, 2019), encompasses a variety of histological subtypes, each presenting unique clinical characteristics. Recognized by the WHO for its complex composition, Combined Small Cell Lung Cancer (C-SCLC) stands out as an uncommon variant (Wagner et al., 2009). Despite its clinical significance, there exists a knowledge gap in the diagnosis, treatment, and prognosis of C-SCLC, particularly when compared to more extensively studied subtypes such as Non-Small Cell Lung Cancer (NSCLC) and Small Cell Lung Cancer (SCLC). C-SCLC represents a unique subset of lung cancer where small cell lung cancer (SCLC) coexists with non-small cell lung cancer (NSCLC) components. This dual histology presents significant implications for diagnosis, treatment, and prognosis. The presence of both small cell and non-small cell elements can complicate the clinical approach, as each type of cancer may respond differently to treatment modalities. For instance, SCLC is typically more aggressive but responds better to chemotherapy and radiotherapy compared to NSCLC (Zeng et al., 2023).

Further insight into the importance of this research is highlighted in this study assessing the differentiation of treatment outcomes among lung cancer types, NSCLC and SCLC including C-SCLC, leveraging the SEER database [cite]. They concluded that C-SCLC has better survival outcome than non-CSCLC in early stage, reinforcing the call for targeted research and clinical approaches (He et al., 2020). Therefore, as pointed out by Dong et al., (2022), the relatively rare type of SCLC, C-SCLC has not been accurately statistically analyzed.

Machine learning has been helpful in assisting healthcare in recent years (Mohanty et al., 2021) and different models are trained for different tasks. However, extensive previous studies using machine learning with clinical data, in the recent years have concentrated on the general prediction or classification of lung cancer survival time, with a specific emphasis on major subtype NSCLC and moving lesser towards the next major subtype SCLC (Tran et al., 2023; Agrawal et al., 2012).

The most crucial aspect of any data-driven research is the quality of data assessed and used. For our study, we use data from the SEER database [cite]. The SEER database is openly available authoritative cancer registry for the United States, encompassing over 48% of the nationwide cancer registries across the nation (Doll et al., 2018).

As years go by, staging systems improve and the algorithms to classify cancers and their characteristics do too, therefore causing discrepancies in the definition of cancer characteristics over time. This causes an issue for longitudinal studies. Machine learning models suffer from sparse data instances and rare cancers such as C-SCLC may need several years' worth of information to create a substantial dataset. To address the existing research gap, in this study, we propose two research questions:

- How effectively can a predictive model forecast the overall survival (OS) months for patients diagnosed with C-SCLC, from the SEER database?
- What are the key predictive clinical factors contributing to overall survival time?

The rest of the paper is structured as follows: In Section 2, we discuss related prior work using ML techniques.

Section 3 presents our methodology including data processing and feature engineering methods. Section 4 presents our predictive models and results. We discuss the insights and implications of our findings in Section 5 and conclude the paper with future work in Section 6.

2. Background and Related Work

Machine learning and analytical studies specifically addressing rare lung cancer types like C-SCLC, have been limited (Li et al., 2023; Huang et al., 2023). Additionally, lesser attention has been given to utilizing the SEER database for in-depth investigations into this rare lung cancer using the “Survival Months” as the variable to predict, our target label feature (Qie et al., 2023; Yang et al., 2021).

A recent review paper highlights numerous gaps in current research when it comes to lung cancer prediction from data-driven approaches. In this review out of the thirty papers analyzed, most of them happened to be from the SEER database. The paper highlights important gaps that we cover in our research (Altuhaifa et al., 2023).

(Altuhaifa et al. 2023) underscores a critical gap in existing research—specifically, the extensive oversight of crucial data preprocessing steps like handling missing data, normalization, scaling, and standardization. These omissions not only introduce potential biases but also compromise the generalizability and accuracy of ML models.

Furthermore, they highlight a consequential disparity in the selection of ML models, noting a lack of uniformity in aligning model choice with the intrinsic characteristics of the dataset. This misalignment often results in suboptimal predictive performance or even overfitted data underscoring the necessity for a more clear and tailored approach to model selection (Altuhaifa et al., 2023).

Li et al., (2023) discuss about the development of medical science that has entered the era of big data, and machine learning has become an essential tool for mining medical big data and acknowledges the SEER database as a wealthy repository of epidemiological data. They highlight important limitations of machine learning such as overfitting and poor interpretability, which need to be weighed in practical applications. This can result in a good fit on the training set but poor performance on the test set. Second, the interpretability of machine learning models is important. Neural network models may have a black-box effect, making it difficult to understand their complex internal functions using interpretable logic, but decision trees could be used to see to a very large extent the decision-making process. Finally, they too highlight the issue of variable selection, as in common with (Altuhaifa et al., 2023), which is a crucial issue in model building, as the inclusion of irrelevant or redundant variables can reduce the model's performance.

Through further exploration, we found other gaps in the robustness of the machine learning applied and used. Especially as seen short of in Huang et al., (2023) through data leaking in their experiment, reviewed by but not observed by Atuhaifa et al. (2023). We offer a thorough assessment of our models, in handling clinical data, which has evolved over the years and is always prone to change (Goldstraw et al., 2007) and get better (Edge et al., 2010; Goldstraw et al., 2016; Amin et al., 2017) as seen in the 8th edition of the AJCC where they claim to make a more “personalized” approach than a “population-based” (Hueman et al., 2017).

The identification of C-SCLC is crucial as it necessitates a tailored treatment strategy that addresses both components (small and non-small) effectively. Moreover, understanding the biological behavior and interaction between the two

cancer types in C-SCLC can provide valuable insights into lung cancer pathogenesis and potential therapeutic targets. We address the omissions in prior work by being mindful of the staging systems used at the time of recording, making the models interpretable, careful implementation to ensure proper fitting of the model (to avoid overfitting), transparent selection of the features, filtering of the variables, etc. Finally, we present our code to reproduce our results. We provide insights into the challenges of low data points and longitudinal studies also addressing the often-overlooked aspect of clinical relevance in lung cancer survival prediction, generalizability. By conducting a detailed examination for generalizability amongst other databases such as the National Cancer Database (NCDB) (Lerro et al., 2013), we contribute knowledge that can be transferred, to assist clinical decision-making and move towards the more “personalized” approach.

3. Methodology

3.1 Data source and selection criteria

The SEER database (The Surveillance, Epidemiology, and End Results program) is a comprehensive source of cancer epidemiology data in the United States (Hankey et al., 1999). Below are the details of the data extracted from the SEER database, using the provided SEER stat tool:

Database name: SEER Research Data, 17 Registries, Years (2004-2020).

Columns extracted : Patient ID, Age recode with <1 year olds, Marital status at diagnosis, Year of diagnosis, Sex, Race recode (W, B, AI, API), Derived AJCC T, 6th ed (2004-2015), Derived AJCC Stage Group, 6th ed (2004-2015), Derived AJCC T, 7th ed (2010-2015), Derived AJCC N, 7th ed (2010-2015), Derived AJCC M, 7th ed (2010-2015), Derived AJCC Stage Group, 7th ed (2010-2015), Derived SEER Combined T (2016-2017), Derived SEER Combined N (2016-2017), Derived SEER Combined M (2016-2017), Derived SEER Cmb Stg Grp (2016-2017), Derived EOD 2018 T (2018+), Derived EOD 2018 N (2018+), Derived EOD 2018 M (2018+), Derived EOD 2018 Stage Group (2018+), Chemotherapy recode (yes, no/unk), Radiation recode, RX Summ--Scope Reg LN Sur (2003+), CS tumor size (2004-2015), Tumor Size Summary (2016+), Survival months.

Year of Diagnosis: Only includes records from 2004 to 2020.

Histologic Type ICD-O-3: Include only cases with code 8045.

Cause of Death: Include cases where death is attributable to the diagnosed cancer.

A total of 1619 patient records were taken into consideration to establish a working dataset.

3.2 Data Understanding and Visualization

Initial exploratory data analysis is crucial in understanding correlations, and making sense of the data. The records of diagnoses from 2004 to 2020 remain relatively consistent each year, indicating stable data collection and no drastic fall in incidence rates over this period.

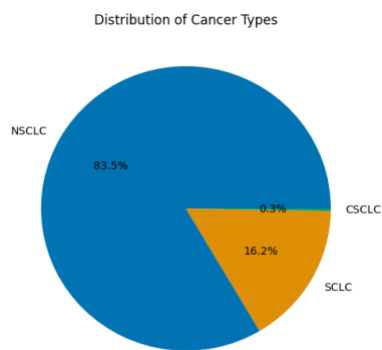


Figure 1: Distribution of Survival Months

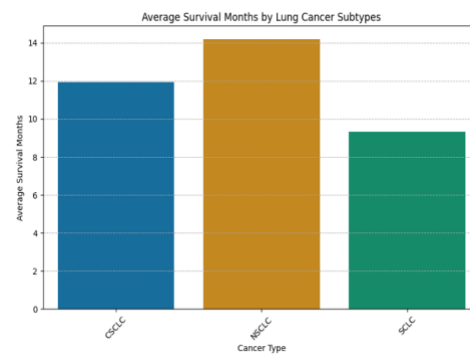


Figure 2: Average Survival Month by Cancer Group

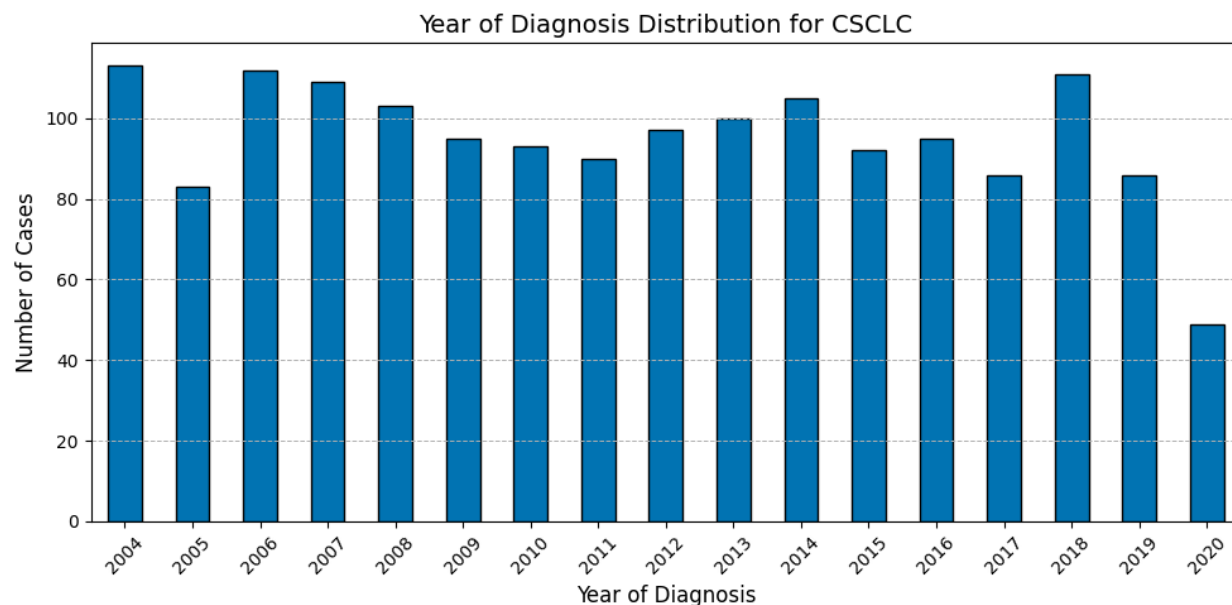


Figure 1: Year of Diagnosis Distribution for CSCLC

It is clear from Figs. 1 and 2, that C-SCLC has the least distribution and smallest survival time. Fig. 3 shows that C-SCLC incidence rate has remained steady over the years.

3.3 Data Preprocessing

Preprocessing the dataset is pivotal in ensuring the integrity and quality of the subsequent analysis. The TNM staging system, introduced in 1940, has undergone numerous revisions to enhance the precision of cancer diagnosis and treatment planning. These revisions reflect advancements in medical understanding and technology, leading to improved survival predictions. However, the dynamic nature of staging criteria, especially between the 6th and 7th editions of the AJCC and the Extent of Disease (EOD) coding, presents a unique challenge for longitudinal studies. Changes in the definitions of T (tumor size and extent), N (nodal involvement), and M (presence of metastases) classifications mean that the same cancer is staged differently across editions.

Some changes include reclassification of T2 tumors that were greater than 7cm (about 2.76 in), that were T3, malignant pleural and pericardial effusions moved from T4 to M1a and to make changes, it requires making certain assumptions and understanding that there may be inconsistencies and limitations when comparing data across different staging editions (American Joint Committee on Cancer [AJCC], 2002, 2010; Lababede and Meziane, 2018). It is essential, as mentioned in their release, that everyone not only documents a stage classification but also the edition used, for example T1bN1M0 (seventh edition)/T1cN1M0 (eighth edition) (Detterbeck, 2018). The EOD 2018, based on the 8th edition AJCC addresses these limitations of earlier versions by ensuring that information on direct tumor extension is preserved. This structured approach allows for a more precise classification of the extent of cancer, facilitating better patient management and contributing to more reliable epidemiological data for research¹

To address these issues, we construct models for the 6th edition only, even though using the latest staging system is usually advisable. Ideally, modeling the latest stage, i.e AJCC 8th or 9th is the most novel approach for reproducibility and usage but the entries for the EOD in this edition hasn't been updated recently and given the rarity of this cancer, we suffer from sparse data points for modeling. The 7th edition was only recorded from (2010-2015) offering a smaller sample size for an outdated stage. This strategy, of using only the 6th edition, allows us to preserve the integrity of each staging system's criteria, providing a more accurate analysis of survival outcomes relative to the specific criteria used

¹ see <https://crgc-cancer.org/wp-content/uploads/2013/09/Not-the-Same-Old-EOD.pdf>

at the time of diagnosis. This choice is due to most available data without missing points, after thorough analysis, for C-SCLC that is between the period 2004 to 2015.

Most of the papers from the recent research on lung cancer prediction have not been clear and lack details about preprocessing (Altuhaifa et al., 2023). The reason could be due to the lack of clear instructions on longitudinal studies from SEER itself in handling different stages and systems variables. However, through meticulous review, we have found more research that harmonizes the TNM stages across staging editions, without the clarity of steps taken to (Cree et al., 2020; Brierley et al., 2017; Stessin et al., 2014; Fan et al., 2022; Jiang et al., 2024; Luo et al., 2023; Momenzadeh et al., 2021; Liu et al., 2022; Liu et al., 2023; Hao et al., 2023). A few pieces of research have shared information on their code and preprocessing (Fan et al., 2012; Yan et al., 2023; Zhou et al., 2023) doing the same as mentioned above. Converting the patients diagnosed with the 6th edition EOD to the 7th to the 8th and now the 9th which is to be released soon, is a non-trivial task, it considers multiple changes and algorithmic differences in assessing them. This is the limitation of the SEER database, and EOD was created to fill this gap.

3.4 Data Preparation for Model Building

Grouping and transformation techniques were applied to refine and categorize the data effectively. Records with ambiguous or non-applicable staging information are excluded. For the age column, grouping lower frequency counts was done to ensure a more uniform distribution of age categories. Marital status was reclassified into simplified groups using a predefined mapping, merging related statuses to reduce complexity and enhance interpretability. Similarly, race categories were streamlined by grouping 'Asian or Pacific Islander' and 'American Indian/Alaska Native' into an 'Other' category. The radiation treatment variable was transformed into a binary indicator, distinguishing between the presence and absence of radiation therapy, while surgical intervention data were dichotomized, distinguishing between known surgical cases and those either unknown or not applicable. Further simplification was achieved by categorizing

TNM staging and Group stage into their main categories because sub-categories like T1a, T1b, and T1c, contributing to the dimensionality challenge, and tumor size was constrained values indicating exact size in millimeters.

Feature Name	Unique Values	Counts	Proportions (%)
Age	25-39 years	2	0.23
	40-44 years	10	1.16
	45-49 years	26	3.01
	50-54 years	51	5.91
	55-59 years	84	9.73
	60-64 years	122	14.14
	65-69 years	166	19.24
	70-74 years	149	17.27
	75-79 years	127	14.72
	80-84 years	91	10.54
Marital Status	85+ years	35	4.06
	Married	483	55.97
	Single	234	27.11
	Single (never married)	112	12.98
Sex	Unknown/Other	34	3.94
	Female	375	43.45
Race	Male	488	56.55
	White	745	86.33
	Black	85	9.85
Chemotherapy	Other	33	3.82
	1 (Yes)	533	64.08
Surgery	0 (No)	310	35.92
	1 (Yes)	287	33.26
Radiation recode	0 (No)	576	66.74
	1 (Yes)	382	44.26
T	0 (No)	481	55.74
	T0	6	0.70
	T1	178	20.63
	T2	302	34.99
	T3	59	6.84
	T4	287	33.26
N	TX	31	3.59
	N0	279	32.33
	N1	100	11.59
	N2	347	40.21
M	N3	105	12.17
	NX	32	3.71
	M0	433	50.17
Grouped Stage	M1	415	48.09
	MX	15	1.74
	I	119	13.79
	II	64	7.42
Tumor Size	III	236	27.35
	IV	426	49.36
	Other	18	2.09
Tumor Size		46.49 (Mean)	

Table 1: Features included in dataset for modeling.

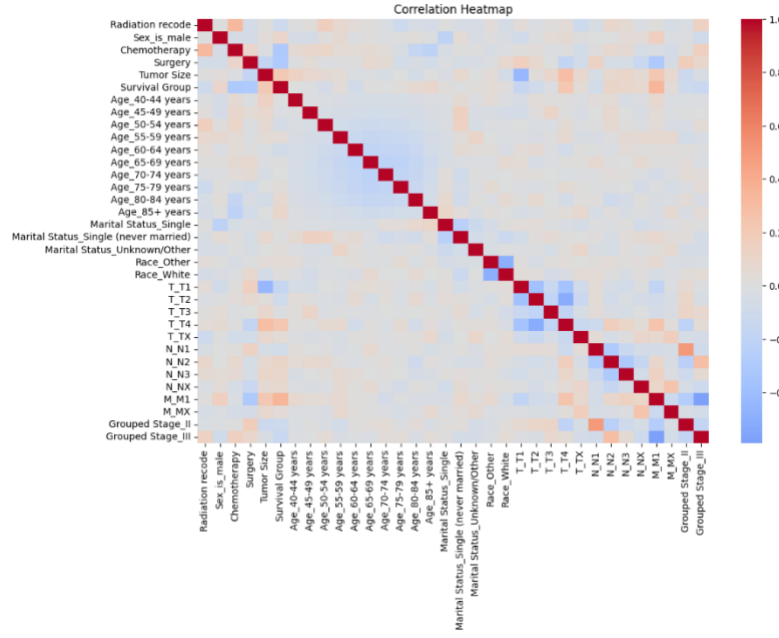


Figure 2: Correlation Heatmap after One-hot Encoded.

To deal with the high correlation of M1 and Stage 4, stage 4 was dropped from modeling as through review of the AJCC guidelines, Stage 4 is any metastatic cancer.

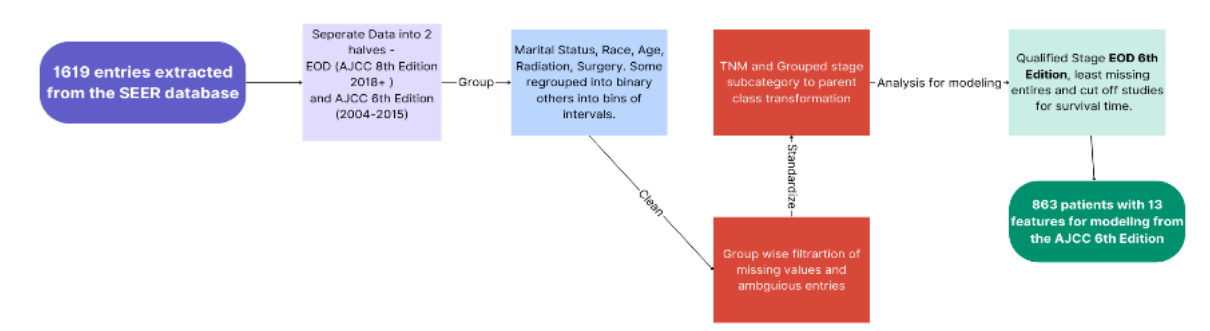


Figure 3: Process Flowchart

We started with 1619 patients and 27 features (all stages from 2004-2020) and ended up with 863 records and 13 features after cleaning and preprocessing, for the years 2004-2016. Our final 13 features from the AJCC 6th edition are - Age, Sex, Race, Marital Status, Radiation recode, T (size and extent of the main tumor), N (lymph node involvement), M (tumor dissemination to other organs), Grouped Stage, Chemotherapy, Tumor Size (Physical dimensions of the tumor in mm), Surgery, with our target variable being Survival Months (see Table 1).

4. Modeling and Results

Past studies (Yang et al. 2022) mentioned about development of a deep survival learning model (DSLML) that outperforms traditional models like Cox proportional hazards (CPH) and random survival forest (RSF) models showing that traditional statistics could not hold as a benchmark with complex non-linear relationships in the data. In our study, we build several advanced machine learning classifiers. We perform a stratified split of the dataset into training and testing sets, with 30% of the data reserved for testing to ensure that the distribution of classes in the split's mirrors that of the original dataset.

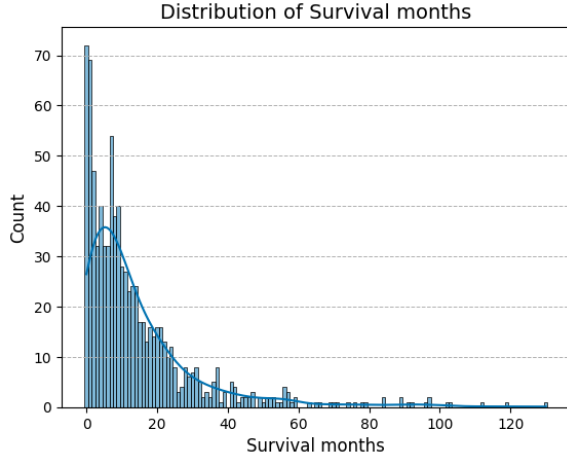


Figure 4: Distribution of Survival Months

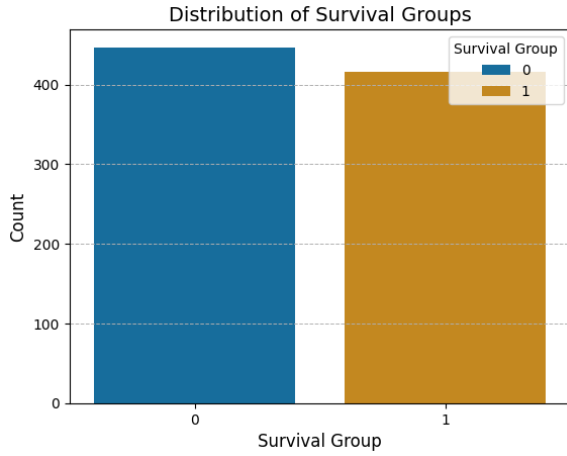


Figure 5: Distribution of Survival Groups

Our target variable, Survival Months, was grouped into two categories, denoted by (0) to be above 9 months survival or below (1), owing to the median of the distribution. Multiple papers follow a similar pattern in the grouping strategy (Mei, 2017; Huang, et al., 2023; Siah et al., 2019) pointing at more medical-backed reasoning that would make models like these more versatile, as discussed in (Bartholomai et al., 2018) which addresses the accuracies in physicians’ judgments on the compared to the real survival time.

We employed Optuna, a well-established hyperparameter optimization framework, to automate the search for optimal hyperparameter values. This allowed us to efficiently explore a large search space and leverage Bayesian optimization for efficient convergence.

Model	Accuracy	Precision for Class 1	Recall for Class 1	F1-score
DecisionTree	0.69	0.68	0.77	0.73
RandomForest	0.71	0.72	0.75	0.73
SVM	0.73	0.72	0.81	0.76
KNN	0.69	0.69	0.76	0.72
LogisticRegression	0.73	0.75	0.74	0.74
LightGBM	0.69	0.70	0.75	0.72
MLP	0.73	0.76	0.72	0.74

Table 2: ML model results showing various performance metrics.

The study tested models such as Decision Tree, RandomForest, SVM, KNN, Logistic Regression, LightGBM, and MLP. This analysis is crucial in identifying models that exhibit superior performance in terms of accuracy, precision, recall, and F1-score for Class 1, which represents the high-risk group.

Among the models assessed, the SVM and Logistic Regression models demonstrated exceptional performance, both securing an accuracy of 0.73. The SVM model stood out with the highest recall for Class 1 at 0.81 and also the highest F-1 score of 0.76. Recall for Class 1 is of paramount importance in our analysis, as it directly relates to the model's effectiveness in detecting all actual positive cases, which, in this context, are patients who are likely to survive less than 9 months. This is critical in oncological prognosis, where early identification of high-risk patients can significantly influence treatment decisions and outcomes.

The inclusion of ROC curve analysis complements our performance metrics, offering a visual and quantitative representation of model efficacy across various threshold settings. The ROC curves of the evaluated models present a summary of their capability to discriminate between the prognostic outcomes. The Logistic Regression model exhibited an AUC of 0.81, indicating a strong predictive power, while the SVM model followed closely with an AUC of 0.79. These AUC curves reflect the models' overall ability to balance true positive and false positive rates and are especially informative when prioritizing the correct identification of high-risk patients.

Furthermore, the Logistic Regression model exhibited a notable balance across metrics, with a precision of 0.75 for Class 1, closely followed by the MLP model, which showed a precision of 0.76 for the same class. Although the MLP model presented a slightly lower recall for Class 1 at 0.72, its balanced precision and F1-score suggest its usefulness in specific scenarios where a more complex interpretation of patient risk is necessary. The Random Forest model, while offering a solid recall of 0.75 for Class 1, also provides a robust option for scenarios where the identification of patients with less than 9 months of survival time is crucial.

The SVM model's superior recall for Class 1 positions it as a preferred choice for prioritizing the detection of high-risk patients, aligning with healthcare objectives that emphasize the reduction of false negatives in patient prognosis.

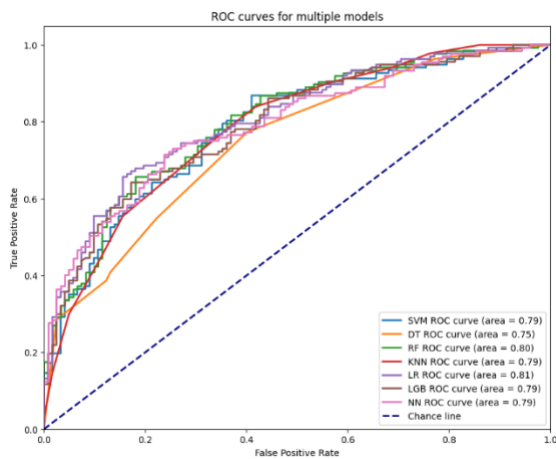


Figure 6: ROC Curves for multiple models.

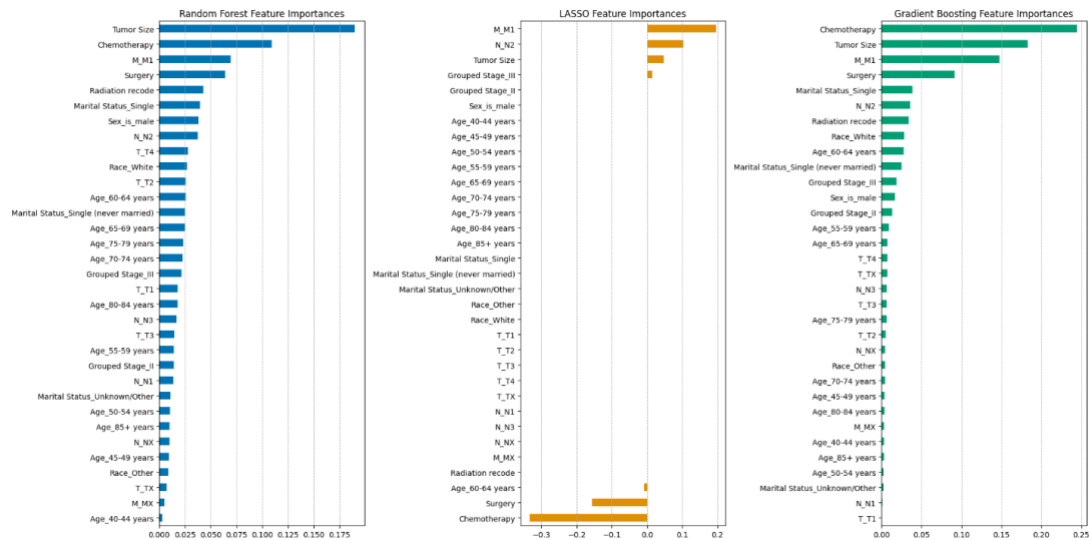


Figure9: Feature Importance in Different Models

In the feature importance diagrams from three different machine learning models—Random Forest, LASSO, and Gradient Boosting—we observe key variables that influence the prediction of survival in C-SCLC patients with a prognosis of less than or greater than 9 months. Tumor size and the presence of metastasis (M1) emerge as the most significant factors across all models, indicating their strong impact on survival outcomes. Chemotherapy and surgical interventions also feature prominently, suggesting that treatments significantly sway prognostic predictions. Marital status, particularly being single, and primary tumor classifications (N2, T4) are relevant in the Random Forest and Gradient Boosting models, reflecting the potential influence of social support and tumor progression on patient survival. Variables related to race, age, and grouped stages of cancer appear with varying degrees of importance, highlighting the multifaceted nature of cancer prognosis. Notably, features like TX, NX, and MX, representing unknown primary tumor, regional lymph nodes, and metastasis status, are deemed less influential in Gradient Boosting.

5. Discussion and Future Work

Our study on Combined Small Cell Lung Cancer (C-SCLC) offers an examination of the challenges and implications of applying machine learning models for cancer prognosis with rare subtypes, for longitudinal studies and inter-stage analysis.

Our contribution made through this study are as follows:

- Representation of a crucial and underrepresented cancer subtype for advanced analysis
- Maintaining integrity in the data by being mindful of the staging systems
- Clear preprocessing, feature selection and engineering steps
- Thorough assessment of models, through train/test/validate splits to avoid overfitting and making models generalizable.
- Making models interpretable through feature importance measures, to ensure that oncologists know exactly the features being considered by the models for prediction
- Using advanced machine learning over traditionally used statistical methods.

This also underscores the importance of external validation across diverse datasets, such as the National Cancer Database (NCDB) (Lerro et al., 2013; She et al., 2020), to ensure the applicability of our findings beyond the initial SEER database context. That is the limitation of this study, although the variables are generalizable across the databases, the physical implementation would need additional access rights and procedures.

To answer our research questions, the SVM (Support Vector Machine), in this setting, brings us the highest recall (81%) with the most important features being Metastasis, Chemotherapy, Radiation, Surgery and Tumor Size.

Our analysis also acknowledges the need for clear guidelines in longitudinal studies, especially for rare cancer subtypes, where small sample sizes necessitate extended observational periods to derive statistically significant insights. To advance this research on the classification and distinguishment of C-SCLC, the incorporation of genomic and molecular data into predictive models would represent a promising avenue to refine prognosis predictions and personalize patient care, to not just rely on a population-based database but take the predictive powers of machine learning to more complex relations and databases. Another missing factor in the SEER database, that is of most importance and the leading cause of Lung Cancer related causalities, is the use of tobacco. Future research will extend beyond the confines of Combined Small Cell Lung Cancer (C-SCLC) to encompass all lung cancer subtypes. We aim to leverage even more sophisticated machine learning techniques, designed to tackle the challenges presented by imbalanced datasets. This approach will ensure that less prevalent cancer types are accurately represented and analyzed, thereby enhancing the predictive models' overall efficacy and applicability.

6. Conclusion

Our study on Combined Small Cell Lung Cancer (C-SCLC) presents advancements in understanding and predicting the disease's outcomes using the SEER database. By focusing on C-SCLC, a variant often sidelined in medical literature despite its complexity and clinical significance, we've applied advanced machine learning techniques to improve survival predictions' accuracy and interpretability.

We also explore the clinical implications of our findings, offering insights that could influence patient care. By identifying key predictive factors, our research supports personalized treatment approaches, which is a current goal of the AJCC as they moved into the 8th edition of their staging systems. This model could identify the patients, through data-driven approaches, that are in most need for urgent care or not with effective reasoning. This could help hospitals or caregivers in resource allocation and prognosis of patients with complex cancer subtypes.

This paper also addresses the major gaps in prediction studies for survivability when it comes to robust techniques or interpretability. Our approach accounts for the evolution in cancer staging, embracing the classifications of the AJCC's 6th edition, which in turn has enhanced the robustness and relevance of our predictive outcomes. The implications of our study are multifold, impacting both the data science community and clinical practice.

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